

125

Erin Buchanan

11:24

Time to complete

Background and Level Selection

NOTE: After you submit the Application, be sure to click the 'Save My Response to Edit' button so you will be able to update or revise your application if needed.

1. Name: *

Erin Buchanan

2. Email: *

ebuchanan@harrisburgu.edu

3. Program *

ANLY

4. Project Title (should be descriptive of the type of research to be conducted): *

Many Languages, One Network: Cross-Linguistic Semantic Priming Effects in 45 Languages and Large Language Models

5. Enter your CITI training Record ID (located in upper right corner of certificate). *

46135148

The value must be a number

6. Upload your CITI training completion certificate. *



buchanan_HU_citi_Erin Buchanan 1.pdf

7. Position at HU: *

- ☐ Undergraduate Student
- ☐ Masters Student
- ☐ PhD Student
- ☒ Faculty/Staff

Faculty Information

8. Names of Co-PIs or collaborators

No answer provided.

9. Emails for Co-PIs/collaborators

No answer provided.

10. Are you seeking funding for this project/research? If yes, please email a copy of the proposal summary/abstract to IRB@harrisburgu.edu *

- ☒ Yes
- ☐ No

11. Does the funding agency require IRB approval?

- ☒ Yes
- ☐ No

12. Does your project involve collecting data on or from individual humans or animals? *

Note. Biospecimens such as cellular data is a type of human data requiring at minimum an Exempt application.

- ☐ No, research will not collect data about or from humans
- ☐ No, research will collect data on humans but is aggregated and deidentified (and does not pose potential harm to groups or individuals)
- ☒ Yes, research will collect data from individuals directly (e.g. survey, interviews, experiments)
- ☐ Yes, data will be collected from animals

Select Application Level

13. What is the expected sample size?

50,000 or more

14. Which of the following types of participants will be used? *

- ☒ Adults (18 years or older)
- ☒ HU Students (18 years or older)
- ☐ Faculty or external collaborators
- ☐ Children (17 years or younger)
- ☐ Patients in Institutions
- ☐ Prisoners
- ☐ Pregnant women
- ☐ Other

15. Which level of application does your study fall into? *

- ☐ Exempt: Research using public data on individuals, using secondary or archival research data on individuals (with some exceptions), research methods and techniques, and some uses of educational data
- ☒ Expedited: Survey research of individuals with no more than minimal risk, Interviews of individuals with no more than minimal risk, Experimental research with no more than minimal risk, or Use of public or secondary where there is more than minimal risk (including physical, social, psychological, and privacy risks for either individuals or groups)
- ☐ Full Board: Any research involving more than minimal risk, the collection of sensitive data on individuals, or protected classes (i.e. children, pregnant people, prisoners, and individuals with mental disabilities or educational disadvantages)

Expedited Applications (review time: up to 2 weeks for initial decision)

Survey research of individuals with no more than minimal risk, Interviews of individuals with no more than minimal risk, Experimental research with no more than minimal risk, or Use of public or secondary where there is more than minimal risk

16. What category of research is involved? See OHRP for further information and criteria, <<https://www.hhs.gov/ohrp/regulations-and-policy/guidance/categories-of-research-expedited-review-procedure-1998/index.html>>. If conducting a survey, interview, or focus group, Category 7 is likely the most appropriate category.

- ☐ Category 1: Clinical studies of drugs and medical devices only when condition (a) or (b) is met,
- ☐ Category 2: Collection of blood samples by finger stick, heel stick, ear stick, or venipuncture from healthy, non-pregnant adults.
- ☐ Category 3: Prospective collection of biological specimens for research purposes by noninvasive means.
- ☐ Category 4: Collection of data through noninvasive procedures (not involving general anesthesia or sedation) routinely employed in clinical practice, excluding procedures involving x-rays or microwaves.
- ☐ Category 5: Research involving materials (data, documents, records, or specimens) that have been collected, or will be collected solely for nonresearch purposes (such as medical treatment or diagnosis).
- ☐ Category 6: Collection of data from voice, video, digital, or image recordings made for research purposes.
- ☒ Category 7: Research on individual or group characteristics or behavior (including, but not limited to, research on perception, cognition, motivation, identity, language, communication, cultural beliefs or practices, and social behavior) or research employing survey, interview, oral history, focus group, program evaluation, human factors evaluation, or quality assurance methodologies.
- ☐ Category 8: Continuing review of research previously approved by the convened IRB.
- ☐ Category 9: Continuing review of research, not conducted under an investigational new drug application or investigational device exemption where categories two (2) through eight (8) do not apply but the IRB has determined and documented at a convened meeting that the research involves no greater than minimal risk and no additional risks have been identified.

17. **Consent Form.** All surveys & interviews must include a consent form with the purpose of the study, expectations of participants, potential risks and benefits, any costs or compensation, and researcher contact information. A template is available on the IRB Sharepoint site (<https://myharrisburgu.sharepoint.com/sites/HUIRBHub>) *

CONSENT FORM

Understanding Word Processing and Meaning

You are invited to be in a research study about how you read and process words, along with their meaning. We ask that you read this form and ask any questions you may have before agreeing to be in the study.

This study is being conducted by Dr. Erin M. Buchanan, Professor of Cognitive Analytics at Harrisburg University of Science and Technology.

Background Information:

In this study, you will be asked to complete different questions about word concepts. For example, you may be asked to define a word's characteristics, rate how familiar you are with a word, or simply judge if a string of letters is a real word.

Procedures:

You will take this study entirely online from a desktop or laptop computer with a keyboard. You will be given instructions about the experiment sections which are randomly selected for each person. After you complete the experiment, you can learn more about the study and goals of the research. The entire study should take less than thirty minutes to complete.

Risks and Benefits of being in the Study:

No identifying information will be collected from you, and therefore, your responses should be anonymous. The current study is similar to an online game, which may cause some fatigue or boredom based on the task you are asked to complete.

There is no direct benefit to you for participating in this study. However, your responses will contribute to our understanding of language and cognitive memory processes.

Compensation:

You may be compensated when taking part in this study through your local researcher.

Confidentiality and Data Sharing:

Measures are taken to ensure that all information you provide will be anonymous. The data from this project will be posted publicly for other researchers to use; however, no data will be directly linked to you. Your name or other identifying information will not be entered into the dataset and no references will be made in verbal or written reports that could link you to the study. In any publication, information will be provided in such a way that you cannot be identified.

Before your data are shared outside the research team, any potentially identifying information will be removed. The anonymous data may be used by the research team or shared with other researchers, for both related and unrelated research purposes in the future. Your anonymous data may also be made available in online data repositories such as the Open Science Framework (which are free data repositories that require registration to have access), which allow other researchers and interested parties to access the data for further analysis.

Please note that your data will be anonymous, which means you cannot ask for it to be removed once you have completed the study.

Voluntary Nature of the Study:

Participation in this study is voluntary:

Your decision whether to participate will not affect your current or future relations with Harrisburg University of Science and Technology or your local institution. If you decide to participate, you are free to not answer any question or withdraw at any time without affecting those relationships.

Contacts and Questions:

The researchers conducting this study are Dr. Erin M. Buchanan in partnership with the Psychological Science Accelerator. You may ask any questions you have now. If you have questions later, you are encouraged to contact Dr. Erin M. Buchanan at ebuchanan@harrisburgu.edu.

Questions or Concerns:

This study has been reviewed by Harrisburg University of Science and Technology's Institutional Review Board (IRB). The IRB has determined that this study fulfills the human research subject protections obligations required by state and federal law and University policies.

A copy of this information to keep for your records will be provided upon request.

18. List all questions that participants will be asked. *

If conducting a survey on QuestionPro, you can provide a link to the survey.

Specific Study Materials:

Demographics:

What is your native (first) language? (this choice will direct them to the full study in their native language).

Please tell us a little bit about yourself.

Please tell us your gender: male female other prefer not to say

Which year were you born? Please enter a four-digit year:

Please tell us your education level: less than High School diploma, High School, Associates or two year degree, University or four year degree, Masters Degree, Doctoral degree (please note these will be made culturally relevant for each language)

Semantic priming task:

This experiment is concerned with how people process words. You will be asked to view words and judge them. The words will be presented in the middle of the screen. You should ask yourself if the word on the screen is a real word or a fake word. If you recognize the word on the screen like COLD, please press the KEY key for real word. If the word is made-up nonsense, like WERM, please press the KEY key for fake word. During practice, we will give you feedback. We would like to ask you to respond as quickly as you can while remaining accurate. You have five seconds to respond to every word. Please press the SPACE BAR to try a few for practice.

(note the KEY will be manipulated between participants and based on the traditional keyboard for that language placing it by the Z and ? keys on a traditional QWERTY keyboard).

Please press KEY for a real word, and KEY for a fake word. (shown while completing the priming task wherein words are shown on the screen)

Great job! Remember you should use the KEY key for real words, and the KEY key for fake words. You will now complete the real task. Remember to go as quickly as possible while getting the correct answer. You will not see any feedback during this section. Please press the SPACE BAR to continue. (shown after practice and before the real trials).

Please take a short break on this screen. You may press the SPACE BAR when you are ready to continue. (shown after each 50-100 pairs to combat fatigue).

Thank you for completing the first part of the experiment. On the next screen, you will judge words for some of their properties to help us measure how these words are understood by native speakers. (a random section below will be shown).

Word meaning task: (McRae et al., 2005)

We want to know how people read words for meaning. Please fill in features of the word that you can think of. Examples of different types of features would be: how it looks, sounds, smells, feels, or tastes; what it is made of; what it is used for; and where it comes from. Here is an example:

duck: is a bird, is an animal, waddles, flies, migrates, lays eggs, quacks, swims, has wings, has a beak, has webbed feet, has feathers, lives in ponds, lives in water, hunted by people, is edible

Complete this questionnaire reasonably quickly, but try to list at least a few properties for each word. Thank you very much for completing this questionnaire.

Age of acquisition task: (Kuperman et al., 2012)

Please indicate (in years) the age at which you learned each of the words on the list. An approximate age is good enough for this rating. If you do not know the meaning of a word, please enter the X key. By "learning a word" we mean the age at which you would have understood that word if somebody had used it in front of you, EVEN IF YOU DID NOT use, read or write it at the time.

Concreteness task: (Brysbaert et al., 2014)

Some words refer to things or actions in reality, which you can experience directly through one of the five senses. We call these words concrete words. Other words refer to meanings that cannot be experienced directly but which we know because the meanings can be defined by other words. These are abstract words. Still other words fall in-between the two extremes, because we can experience them to some extent and in addition we rely on language to understand them. We want you to indicate how concrete the meaning of

19. If applicable, describe any stimuli or materials participants will be shown.

Materials can also be emailed to the IRB (IRB@harrisburgu.edu)

A small demonstration of the experiment can be found at: [here](#). The study will be programmed using lab.js (Henninger, Shevchenko, Mertens, Kieslich, & Hilbig, 2019), which is an online, open-source study creation project. Precise timing measurement is required for this study, and the lab.js team has documented the accuracy of measurement within their framework (Henninger, Shevchenko, Mertens, Kieslich, & Hilbig, 2018), and previous work has shown no differences between lab and web-based data collection for response latencies (Hilbig, 2016). In addition, SPALEX, a large lexical decision database in Spanish was collected completely online (Aguasvivas et al., 2018). We will recommend that research labs use Chrome as their browser, however, meta-information about the browser and operating system are saved when participants take the experiment to control for implementation differences. Participants will be directed to an online web portal to take the study, and all data will be retained in the online platform with nightly backups to GitHub. They will be asked to indicate their gender (male, female, other, prefer not to say), year of birth for age, and education level (none, elementary school, high school, bachelors, masters, doctorate) for demographic variables. To continue in the study, they will select their primary language, which will direct them to the appropriate stimuli set. The research lab ID will be collected (to track external collaborators) but no identifying information will be collected about participants (i.e., IP address or other information than listed here for demographics).

Participants will be required to complete the study on a computer, rather than a mobile or tablet device. This requirement allows for tracking of the display of the device which will indicate important aspects about screen size, browser, and timing accuracy. In order to enforce this requirement, participants will be asked to hit the spacebar to continue the study. Instructions on how to complete a lexical decision task will be shown on the next screen, followed by 10 practice trials. Each trial starts with a fixation cross (+) in the middle of the screen for 500 ms. The concept will then be displayed in the middle of the screen in uppercase San-Serif font (i.e., NURSE). On the bottom of the screen the answer choices will be shown as the traditional keys next to the shift key depending on the common keyboard layout for that language (i.e., Z and / on a QWERTY keyboard or > and - on a QWERTZ keyboard). These choices will be reversed in half of the subjects, which will be randomly selected at the start of the study to counterbalance word/nonword selection. Participants will enter their choice for each concept, and then the next word will appear with an intertrial interval of 500 ms (i.e., the time between the offset of the first concept and onset of the next concept, when the fixation cross is showing). Responses will time out after 5 seconds and move on to the next trial. After ten trials, participants will see the instruction screen again with a reminder that they will now be doing the real task.

After 100 trials, the participants will be shown a short break screen with the option to continue by hitting the spacebar after 10 seconds. After six blocks of 100 trials (600 words), the experiment will end with a thank you screen. On this screen, participants will indicate what type of credit they are receiving for the study (course credit, payment), and they will be given instructions on how to indicate they have completed the study to the appropriate lab. Participants will be allowed to take the study multiple times (see below). These values will be customized based on data collection type (i.e., Mechanical Turk, participant pool, etc.). An estimate for the amount of time required for the study is approximately twenty to thirty minutes including practice trials, instructions, and breaks. We will pilot test the number of stimuli to keep t

20. How will participants be identified and recruited? *

Participants will be recruited through the newly formed Many Languages network, along with the Psychological Science Accelerator.

21. How will be data be secured? *

As part of the grant project, we will build an online app for data collection and storage. The raw data will be encrypted and stored in this online server in accordance with data protection rules via GDPR. Only the lead team will have access to raw data (PI, PhD, and programmer). All other data is displayed/downloaded/etc. via deidentified means.

22. Will any personally identifiable data be collected? If so, why is it necessary? *

No personally identifiable information will be collected as part of the study - age and handedness are required to ensure compliance with age of consent and the study parameters (i.e., how the study is shown to participants). Participants can optionally give information about gender and education level.

We will collect emails as part of the online app for incentive payments and the ability to allow participants to ask for their data to removed. This information never leaves the application.

Submission Complete

Your application has been submitted. Please allow up to 2 weeks for review. An IRB official will be in contact if there are questions or revisions needed prior to granting approval.

NOTE: On the next page, be sure to click the 'Save My Response to Edit' button so you will be able to update or revise your application if needed.